



# Investigation report

## Grounding on Boolarong Bank, Merrandi Channel, Kanooka Heads - Standard investigation

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INVESTIGATION WORKFLOW SUITE

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## Executive summary

On 07/07/2026, the passenger ferry MV Tarrakin Star grounded on Boolarong Bank in the Merrandi Channel at Kanooka Heads. She was inbound on service TF18 in darkness, carrying 38 passengers, a master, a deckhand and a kiosk attendant. At 18:12:04 she struck the bank at 6.8 kt, and she came to rest 54 m outside the dredged channel on a falling tide.

No one was seriously injured. One passenger reported a grazed shin, and two were treated at the wharf for distress. The port workboat had all passengers ashore by 19:26. The vessel herself was damaged. A 180 mm split holed the starboard hull, and water entered the No. 2 void. The starboard propeller, shaft, rudder stock and skeg were also damaged. She refloated at 00:20 and was withdrawn from service, and the channel was closed to commercial traffic for 19 hours.

While the chart plotter worked, the vessel held the channel centreline. At 18:07:06, however, the plotter display froze, and it did not recover. From that second the master eased the heading five degrees to starboard without deciding to turn, and nothing on the bridge showed him the change. He then continued at about eight knots with no means of fixing the vessel's position, although stopping was available at no cost until about 18:11:40. Mangroves on Merrandi Islet screened the No. 4 beacon from the channel between 0.84 nm and 0.34 nm north of it. That was likely to have contributed, because the beacon was the only reference left and it was hidden while the new heading was held. The vessel left the dredged channel on the bank side. The shallow alarm was still at its 2019 setting of 0.9 m under the keel, so it sounded only 16 seconds before contact. That gave 58 m of warning, against a crash stop distance of 92 m.

Four safety issues contributed to the grounding.

- The operator's passage plan named the plotter as the primary means of fixing. Yet it had no night version and no hold point for losing the plotter. That absence very likely contributed.
- The last service was moved into darkness as a timetable change only. The daylight-only route risk assessment, which called for a night passage plan, was not reopened. This likely contributed.
- The defect procedure never required a fault that had cleared itself to be recorded. As a result, a year of plotter freezes never reached anyone ashore, and the firmware fix of 11/2024 was never applied. This likely contributed, although how quickly a recorded freeze would have led to the fix could not be established.
- At three works rounds, the Port Authority deferred clearing the beacon's sightline as unfunded. This very likely contributed.

The investigation examined four further safety issues. None was found to have contributed, but each increased risk. First, the aids to navigation code required no notice or availability record for beacons off declared channels. Second, a solar conversion halved the beacon's nominal range, and no channel user was told of it. Third, the winter roster placed the darkness transit at the end of a split duty, which finished 20 minutes inside the operator's spread limit. Fourth, a counter fitted in 2019 without re-survey blocked the forward saloon lifejacket locker, and the operator reported the same counter on two other vessels.

Two questions remain open. The investigation could not determine whether boat ramp floodlights behind the beacon delayed the sighting, or whether the master was fatigued. After the grounding, by contrast, the master's decision not to refloat a holed vessel reduced risk.

Several responses have followed. The Merrandi Islet canopy was trimmed on 24/07/2026 and placed on a three-yearly cycle. Notice LNM-2026-14 gave the beacon's corrected range, and the aids register was made available to operators. The operator reissued its passage plan appendix.

The investigation recommended a night passage plan with a hold point that requires the way to be taken off the vessel on loss of the primary fix. It also recommended replacing the services agreement's 50 m vegetation scope with a sightline test.

## Event overview

On 07/07/2026, MV Tarrakin Star was operating Tarrakin Ferries' service TF18 from Muntari Island to the town wharf at Kanooka Heads. She is a 24.4 m aluminium catamaran passenger ferry, certificated for 96 passengers. TF18 was the last service of the day and one of the winter timetable's evening returns. Those returns had been made in darkness since 01/06/2026, and this was the thirty-first of them.

The crew comprised a master, a deckhand and a kiosk attendant. The master had been a master for 14 years, 11 of them on this route, and he estimated that he had made about 4,100 transits of the Merrandi Channel. He was on day nine of a twelve-day tour and was working a split duty. The night before, he had slept 6 h 15.

He took over the vessel from the morning master at the town wharf at about 13:05, and both masters gave the same account of the changeover. The outgoing master said that the chart plotter had frozen twice that morning, recovering by itself both times. The relieving master replied that it had done the same to him the month before. The plotter's event log recorded those freezes at 10:42 and 11:17, each followed by an automatic reset. Yet neither master made an entry in the wheelhouse defect book, which held no entry of any kind for 07/07/2026 before the occurrence. Both the master and the deckhand described the afternoon's two round trips as ordinary.

Sunset was at 17:02, and civil twilight ended at 17:29. TF18 departed Muntari Island at 17:35:02 with 38 passengers embarked. The master reported the departure to Kanooka Heads Port Control on VHF channel 12, and Port Control replied that there was nothing to affect the vessel. He described the conditions as clear, with a light westerly and no sea in the entrance.

At 18:03:58 the vessel entered the Merrandi Channel from the north, abeam the No. 1 beacon. The deckhand was on the bridge and asked "Want me to stay up?" The master replied "Nah, go and do the fares, I'll give you a yell coming up on the wharf." From then on, the master was alone on the bridge. Since entering the channel the vessel had held the centreline to within 3 m. At 18:05:30 she was abeam the No. 2 beacon, 1 m west of the charted centreline, making 8.4 kt on a course of 195 degrees, and the master was heard to say "Two ... good."

At 18:07:06 the plotter's event log recorded that the display process had stopped responding. This time no reset followed. The chart picture stayed on the screen with the vessel symbol stationary, and the display would not accept touch input. The bridge audio recorded "Oh, come on." at 18:07:11. At 18:07:26 it recorded sounds consistent with repeated screen taps, and at 18:07:44 the words "frozen again". By then the vessel was 0.68 nm north of the No. 4 beacon, a starboard-hand mark on the western side of the channel.

From the time of the freeze, the vessel's heading began to increase. Over about a minute it moved from 195 to 200 degrees, and it was then held there. The recorded rate of turn peaked at 6 degrees per minute at 18:07:40 and had returned to zero by 18:08:50. Nothing in the machinery accounts for the change. The engines ran steadily at 1,780 and 1,784 rpm throughout, with no throttle movement, steering pressure stayed within its normal band, and the rudder followed helm demand. In interview, the master said that he had eased the vessel five degrees to starboard while he was dealing with the plotter and then looking for the beacon. In his words, "I never made that decision."

At 18:08:31 the master said "Right. Where's four." The vessel was then 12 m west of the centreline and 0.47 nm north of the beacon. At about 18:09 she crossed the western edge of the dredged channel, 20 m west of the centreline. With the new heading held, she was setting west at about 20 m a minute.

At 18:09:40 the audio recorded a sound consistent with a window or door being slid open. The master later explained that he had opened the wheelhouse window in case the glass was the

problem. Looking for the beacon, he said, was "looking into a wall of light", because the town and the car park floodlights formed one glow.

At 18:10:02 the master said "There's four. Bring her back." The vessel was then 0.25 nm north of the beacon and 39 m west of the centreline, outside the dredged channel. The master said that the beacon appeared dead ahead rather than on the starboard bow, and that he knew at once he was west of the line. He believed, however, that he was "just outside the edge, a boat width maybe". So rather than stopping, he steered back to port. The course over the ground eased from 200 back to 197 degrees, which slowed the westward movement but did not stop it. By 18:11:00 the vessel was 58 m west of the centreline and over Boolarong Bank. Her speed had fallen to 7.6 kt, with no throttle movement.

The echo sounder's shallow alarm had been set at 0.9 m under the keel since the unit was commissioned in 2019. It sounded at 18:11:48, when the vessel was 70 m west of the centreline at 7.1 kt. The master said "No no no." At 18:11:52 he moved both throttles to astern together.

Astern thrust was still building, and had not reached full, when the vessel struck Boolarong Bank at 18:12:04 at 6.8 kt over the ground. Grinding and vibration continued for about 9 seconds. She stopped within 11 m and came to rest heeled 4 degrees to starboard on a falling tide. Her position was 74 m west of the charted centreline, 54 m outside the dredged channel and 34 m west of the No. 4 beacon. In the 16 seconds between the alarm and the contact, the vessel covered 58 m. Her trials figure for a crash stop from 8 kt is 92 m.

The master stopped both engines at 18:12:31 and did not restart them. At 18:12:40 he told the passengers over the public address that the vessel had touched the bottom and was stopped, and he asked them to stay in their seats. At 18:13:04 he broadcast a pan pan on VHF channel 16. He gave the position and the persons on board, reported no injuries and no pollution, and stated "I am not attempting to refloat." Port Control acknowledged and called out the port workboat. The master later explained his decision. On a falling tide over an unknown bottom, he judged the vessel safer on the bank than afloat and possibly holed, because "On the bank she can't sink."

At 18:14:52 the master asked everyone to put on a lifejacket. In the aft saloon, passengers opened the locker themselves. The forward saloon was different, because its locker was behind the kiosk servery counter and three passengers were standing at the counter. The deckhand had to go around the end of the counter and hand the lifejackets across it. By about 18:22, all 38 passengers were in lifejackets.

At 18:16 the deckhand sounded the No. 2 starboard void and found water in it. The master reported this to Port Control as slow ingress of approximately 75 mm, and from then on the void was sounded every ten minutes.

The workboat came alongside the port quarter at 18:41. The last passenger was ashore at 19:26, 74 minutes after the contact. No one was seriously injured. One passenger reported a grazed shin, and two were treated at the wharf for distress.

The vessel refloated on the rising tide at 00:20 on 08/07/2026, with about 260 mm of water in the No. 2 void, and was towed to the town wharf. The damage was confined to the starboard side. There, the hull bottom plating was set up over 4.1 m with a 180 mm split, the propeller blades and shaft were bent, the rudder stock was distorted 3 degrees and the skeg was torn away. The port hull, propulsion and steering were undamaged. The vessel was withdrawn from service, and the Merrandi Channel was closed to commercial traffic for 19 hours.

# Analysis

## Introduction

The vessel held the charted centreline for as long as the chart plotter worked. Within the second the plotter display froze, the heading began to change. The vessel continued at transit speed with no means of fixing its position until the shallow alarm sounded, 16 seconds before contact. This analysis follows those events from the bridge out to the equipment, the procedures and the organisations that shaped what the master had to work with.

## People

### The heading change after the plotter froze

From 18:07:06, the second the plotter froze, the vessel's heading increased steadily from 195 to 200 degrees over about a minute and was then held. The recorded rate of turn peaked at 6 degrees per minute at 18:07:40 and was back to zero by 18:08:50. Rudder angle followed helm demand within 1.5 degrees throughout the transit, so the turn was steered rather than imposed on the vessel.

The timing places most of the turn at the plotter. The bridge audio recorded sounds consistent with repeated screen taps at 18:07:26 and the words "frozen again" at 18:07:44, and the rate of turn peaked between the two. The master's spoken search for the No. 4 beacon did not begin until 18:08:31, when the turn was almost complete. It was a turn that went unnoticed rather than one made while looking for the light.

The master's account agrees. In interview he said: "I eased her to starboard while I was fiddling with the plotter and then looking for four. Five degrees. I never made that decision." His safety report says the same.

Nothing on the bridge showed him the new heading. The plotter's vessel symbol was frozen, and the beacon that would have shown the vessel drawing west was screened over that stretch, as discussed under Environment.

No other mechanism accounts for the westward movement. The wind was offshore at 8 kt and there is no cross-set in the channel at that state of tide. Held at 200 degrees against the channel's 195, the vessel moved west at about 20 m a minute and covered the 20 m from the centreline to the western edge by about 18:09.

It is virtually certain that the master eased the heading five degrees to starboard without a decision to turn, and virtually certain that this heading change contributed to the occurrence by taking the vessel out of the dredged channel.

### Continuing at transit speed without a position fix

The AIS track shows no reduction of speed after the freeze. The vessel was making 8.4 kt at 18:07:06 and 8.1 kt at 18:08:00. Its speed fell to 7.1 kt by 18:11:48, but the engine recorder shows no throttle movement until 18:11:52, and the fall is consistent with the increasing drag of shoaling water beneath the hulls. The bridge audio records the master's attention moving to the beacon at 18:08:31, 85 seconds after the freeze, and no consideration of stopping at any point.

The master did not consider stopping when the plotter froze. He went straight to the visual transit because, in his mind, the visual transit is the backup.

At 18:10:02 he saw the beacon dead ahead and knew he was west of the line. He believed he was only just outside the edge, when he was 39 m west of the centreline. He steered back to port rather

than stopping because, on his account, stopping on the edge of a bank he could not see felt worse than easing back in. The course over the ground came back from 200 to 197 degrees. That slowed the westward movement but did not stop it, because the course stayed west of 195 degrees until contact.

Stopping was available throughout. There was no other traffic, the water was flat, and the wind was offshore at 8 kt. The vessel's trials figure for a crash stop from 8 kt is 92 m.

The vessel crossed the western edge 114 seconds after the freeze, and a stop at any point in those 114 seconds would have left it in the channel. A crash stop begun before about 18:11:40 would still have brought it to rest short of the contact point. That window included the 98 seconds after the master knew he was off the line, and a stop then would have left the vessel on the margin of the bank, upright and not holed.

No lookout was on the bridge during this period. The deckhand had been sent to collect fares at 18:04:14, which is normal practice on the route. Nothing in OPS-3 addresses that practice, and the investigation did not test it as a separate factor.

This action sits downstream of a passage plan with no hold point for the loss of the primary position fix, and of a beacon screened where it was needed. Both are analysed below.

It is virtually certain that the transit continued at transit speed from the plotter freeze until the shallow alarm, and extremely likely that this contributed to the occurrence, because a stop at any point before about 18:11:40 would have prevented the contact.

### **The departure from the dredged channel**

The vessel was 1 m west of the charted centreline abeam the No. 2 beacon at 18:05:30 and held that line until 18:07:06. It reached the western edge at about 18:09, 0.40 nm north of the No. 4 beacon. By 18:10:02 it was 39 m west of the centreline, and it was over Boolarong Bank from about 18:11:00. The surveyed grounding position and the furrows in the bed corroborate the recorded track.

The Merrandi Channel is 40 m wide and dredged to 3.2 m. The bank on its western side dries in places and shoals at about 1 in 70 outside the western edge. The grounding follows directly from the vessel being outside the dredged channel.

It is virtually certain that the vessel left the dredged channel on the bank side, and virtually certain that this contributed to the occurrence.

### **The response after the grounding**

The master's actions after the contact reduced the risk to the passengers. Both engines were stopped at 18:12:31, 27 seconds after contact, and were not restarted.

The master addressed the passengers over the public address at 18:12:40, 36 seconds after the impact. At 18:13:04 he broadcast a pan pan on VHF channel 16, passing the position, the persons on board and the absence of injuries and pollution, and stating "I am not attempting to refloat."

He then directed the deckhand to sound the No. 2 starboard void. At 18:17:05 he reported to Port Control "approximately seventy five millimetres on the sounding rod, being monitored ten minutely". Lifejackets were issued to all 38 passengers.

The survey later confirmed a 180 mm split in the starboard hull plating in way of that void. Ingress had reached approximately 260 mm by the time the vessel refloated on the tide at 00:20. The master's reasoning in interview rested on the falling tide, the unknown bottom, the number of people aboard, and a vessel that could not sink while it was on the bank.

The breach was in the compartment the vessel was sitting on. Backing off into 3.2 m of water, at night, on a falling tide, with the ingress unassessed and 38 passengers aboard, would have put a flooding compartment into deep water.

It is virtually certain that the master stopped the engines and made no attempt to refloat, and very likely that this decision reduced the risk to those on board.

### **The verbal handover of the plotter fault**

At the change of master at about 13:05 at the town wharf, the outgoing master told the relieving master that the plotter had frozen twice that morning and had recovered itself both times. The plotter's log recorded those events at 10:42 and 11:17. The relieving master replied that it had done the same to him the month before.

Neither master entered the fault in the wheelhouse defect book or reported it to the Operations Manager. The book held no entry of any kind for 07/07/2026 before the occurrence. Both accounts of the handover agree in every particular, and both men volunteered the omission before being asked.

This omission did not influence the occurrence. An entry made at 13:05 would have been reviewed by the Operations Manager at the end of the week. Nothing written that afternoon could have applied firmware to the unit, or taken the vessel off TF18, before 18:07.

The omission still matters because it reflects practice rather than a lapse. Both masters described the same rule, the product of the defect procedure analysed under Procedures. Under it, a cleared fault on the primary means of position fixing passes from master to master by word of mouth while nobody ashore learns of it.

It is virtually certain that the plotter fault was passed on verbally and not entered in the defect book. It is unlikely that this contributed to the occurrence, and likely that it increased risk.

### **Whether the master was fatigued**

The investigation examined whether the master was fatigued during the channel transit and could not settle the question. It is about as likely as not that he was. The matter remains unresolved and is not a finding either way.

The evidence in favour is substantial:

- The master was on day nine of a twelve-day tour, on a split duty of 05:40 to 08:40 and 13:00 to 19:20.
- That duty had a spread of 13 hours 40 minutes, with the channel transit falling in its last hour and a quarter.
- His fatigue diary gave 6 hours 15 minutes of sleep the night before.
- In interview he said "I was tired. I'd be lying if I said otherwise."
- Having seen the beacon dead ahead, he eased back to port rather than stopping, believing he was just outside the edge when he was 39 m west of the centreline. That is the kind of misjudgement fatigue is expected to produce.

The evidence against is equally direct. In the same answer the master said "I wasn't fighting to stay awake and I wasn't slow". His recorded performance agrees:

- His helm work held the centreline to within 3 m for the first two minutes in the channel.
- His response to the shallow alarm took 4 seconds.
- His radio work after the grounding was complete and correct at the first attempt.
- His decision not to refloat was made quickly and was right.

He described 6 and a quarter hours as normal for him on a split duty. Nothing else bears on his state that night.

No detailed fatigue assessment was made, because the investigation judged that one could not change the analysis. Each decision that mattered on the night is either accounted for by another factor, in particular the absence of a hold point, or is contradicted by the recorded data. The roster is analysed under Organisation.

## Environment

### Conditions on the night

TF18 departed Muntari Island at 17:35:02. Sunset was at 17:02 and civil twilight ended at 17:29, so the channel transit was made in full darkness with no moon. Visibility exceeded 10 km and the wind was from 305 to 310 degrees at 7 to 9 kt. The port workboat reported the sea state in the channel at 18:45 as slight, with wind ripple only.

Visibility was not reduced in the sense used by OPS-3 3.6. However, every reference available to the master once the plotter froze was an artificial light.

### The screened No. 4 beacon

A range and visibility check from the port workboat on 18/07/2026, on the inbound track at the same time of night, found that the No. 4 beacon cannot be seen from the channel between 0.84 nm and 0.34 nm north of it.

Over that stretch the sightline passes through the mangrove canopy on Merrandi Islet. The islet stands about 240 m up-channel of the beacon, in the shoal strip between the dredged western edge and the bank, and is about 10 m across. Its canopy measured 5.8 m against a lantern height of 4.2 m above chart datum. Beyond 0.84 nm the sightline passes west of the islet, and inside 0.34 nm it passes east of it.

The Port Authority's own register had recorded this screening since 09/2023, and in 03/2026 recorded the sightline from the northern approach as fully closed. Both masters raised the beacon unprompted. Neither had ever reported it.

With the plotter frozen, the beacon was the only means of position fixing left, and it was screened while the new heading was held. The bridge audio has "Right. Where's four" at 18:08:31 and "There's four. Bring her back" at 18:10:02, 91 seconds apart. The vessel was 0.47 nm north of the beacon when the master began looking. It crossed the western edge at about 0.40 nm, inside the screened arc, and was 0.25 nm north when he called the light. A light visible on the starboard bow, where it belonged, would have shown the vessel drawing west within a minute of the turn, while it was still in the channel.

Two points weigh against this. First, the heading change began at 18:07:06, before the search, so the beacon's absence cannot have caused the turn. It can only explain why the turn went unnoticed. Second, the light cleared the islet at 0.34 nm, at about 18:09:25, at least 37 seconds before the master called it. Because the vessel was west of the inbound track on which the check was made, it would have opened earlier still from the vessel's position. For part of the search there was a light to see and it was not seen, which is the subject of the floodlight question below.

The investigation resolved these points as follows. The turn was made at the plotter, but it was held and went unnoticed because the one reference that would have shown it was screened over the minute and a half that mattered. The final 37 seconds do not undo the screening before them.

It is virtually certain that the No. 4 beacon was screened from the channel between 0.84 nm and 0.34 nm north of it, and likely that this contributed to the occurrence by allowing the held heading to go unnoticed.

### The boat ramp floodlights

The investigation examined whether the floodlights at the new boat ramp affected the master's ability to see the beacon and could not settle the question. It is about as likely as not that they did. The matter remains unresolved and is not a finding.

A survey from the port workboat on 18/07/2026 found 14 unshielded LED floodlights on 12 m columns at the boat ramp and car park, each presenting about 4,800 candela toward the channel. The installation was commissioned on 22/05/2026, ten days before the winter timetable began.

From 0.5 nm north of the No. 4 beacon, the array lies behind the beacon within 4 degrees of the same bearing and within 1.5 degrees of the same elevation. The array is continuously lit, while the beacon flashes green for 0.4 seconds every 2.5 seconds. The Council's planning unit approved the development. The Port Authority's aids to navigation officer was not consulted, and no assessment of the lighting against the channel's aids was required or carried out.

The case that the floodlights mattered rests on two strands:

- **The masters' accounts.** The master described looking into a wall of light, and called the beacon "a nothing" against the town and the new car park. The outgoing master, independently, called it "washed out altogether from up the top".
- **The final 37 seconds.** For at least 37 seconds the beacon was clear of the islet and still not seen. Over that stretch, the floodlights rather than the canopy stood between the master and the light.

The case against has four parts:

- Over most of the search the canopy screened the beacon, and it would not have been visible against a dark background either.
- The heading change began before the search, so the floodlights cannot have caused the turn.
- No measurement exists of what could be seen on the night, only the survey eleven days later. Nothing establishes whether a 0.4 second green flash can be detected against that background.
- The master himself said twice that he could not tell whether the lights were the reason he could not see the beacon.

The remaining question is whether the floodlights delayed the sighting after the light cleared the islet, and so delayed the moment the held heading was noticed. Answering it would require a measure of what was visible from the vessel's track on the night, or an established detection threshold for this flash character against this background. Neither exists.

What is established is that a continuously lit array of this strength sits behind a navigation light, on the bearing from which vessels approach it, without any assessment against the channel's aids.

## Equipment

### The chart plotter freeze

Three independent sources place the freeze at 18:07:06. The plotter's internal event log recorded the display process not responding at that time, with no accompanying reset. The bridge audio matches to the second: "Oh, come on." at 18:07:11, sounds consistent with repeated screen taps at

18:07:26, and "frozen again" at 18:07:44. The master's account is that the chart picture remained with the vessel symbol stationary, and that the display would not accept touch input.

The same cause code, 0x51, had been logged at 10:42 and 11:17 that morning, when the unit reset and recovered itself. In the evening it did not.

An approved electronics contractor bench tested the unit for 46 hours on 21/07/2026, using the same antenna and cabling. The fault did not recur and no defect was found. That result goes to the cause of the freeze, not to whether it occurred. The contractor's report notes that cause code 0x51 is known to the manufacturer and is addressed by firmware issued in 11/2024. The unit was running firmware issued in 03/2024 and had never had that update applied, which is analysed under Procedures.

The passage plan named the plotter as the primary means of position fixing. For the 85 seconds before the search for the beacon began, the master had no position fix of any kind.

The secondary means in the plan was the visual transit of the No. 4 beacon with the town wharf leading lights. The leading lights lie about 1.9 nm beyond the No. 6 beacon and do not resolve as two lights until a vessel is inside the channel's southern third. Where the plotter was lost, the secondary means was therefore the beacon alone, and the beacon was screened.

It is virtually certain that the chart plotter display froze during the channel transit and did not recover, and extremely likely that this contributed to the occurrence.

### **The echo sounder shallow alarm setting**

The echo sounder's alarm log states:

- "Shallow alarm set to 0.9 m under the keel. This is the value written into the setup when the unit was commissioned in 2019 and it has not been changed. The alarm sounded once on the voyage, at 08:11:48 UTC, 16 seconds before contact." [Echo sounder alarm log and setting]

That is 18:11:48 local time, and the bridge audio records the alarm tone beginning at the same second. Both throttles moved to astern at 18:11:52, and contact followed at 18:12:04.

At 7.1 kt the vessel covered 58 m in those 16 seconds, against a trials crash stop distance of 92 m. Astern thrust was developing at contact but had not reached full. With the bank gradient at about 1 in 70, the alarm could not have sounded appreciably earlier with a 0.9 m setting.

The alarm did exactly what it was set to do. The problem lies in the setting relative to the vessel's speed and the shape of the bank. A setting of about 1.6 m under the keel would have sounded about 107 m short of the contact, allowing a crash stop with four seconds to react, and still would not have sounded in the dredged channel at low water. A setting high enough to give forty seconds' warning would sound in the dredged channel at low water. The setting therefore needs to be chosen for this channel rather than simply raised.

It is virtually certain that the shallow alarm, at its 2019 setting of 0.9 m, gave sixteen seconds' warning, and extremely likely that this contributed to the occurrence, because sixteen seconds was less than the time the vessel needed to stop.

### **The forward saloon lifejacket locker**

The deckhand reported that, while issuing lifejackets, she had to ask three passengers to move away from the kiosk counter and pass behind it through a 0.6 m gap to reach the forward saloon locker, which holds 40 adult lifejackets.

A survey on 16/07/2026 found that the locker face cannot be reached from 22 of the 34 forward saloon seats without passing behind the servery counter. With people standing at the counter, the

inboard gap is the only route. The muster placard directs passengers to a locker they cannot see from their seats.

The arrangement approved at build in 2011 had an open bench in that position, with the locker face clear. The counter was fitted during a 2019 interior refit under a minor modification notice, which did not trigger a re-survey of the safety equipment arrangement. The operator has stated that the same counter is fitted to its two other 24 m vessels.

The locker had no bearing on the grounding. All 38 passengers were in lifejackets within about eight minutes of the call at 18:14:52, and all were ashore within 74 minutes of the contact. The vessel was aground, upright and stable, and the muster was unhurried. A fire, a risk of capsizing, a night evacuation into the water, or a deckhand injured in the impact would each make a locker that passengers cannot reach without crew help critical. The arrangement remains in place on this vessel.

It is virtually certain that the forward saloon lifejacket locker was obstructed by the counter fitted in 2019. It is unlikely that this contributed to the occurrence, and very likely that it increased risk.

## Procedures

### The passage plan and the loss of the primary fix

OPS-3 sets out how the channel transit is to be made:

- "3.4 The Merrandi Channel transit will be made on the charted centreline, monitored on the chart plotter, with the No. 2 and No. 4 beacons identified visually before the vessel passes the No. 1 beacon." [Operations Manual OPS-3]

The Merrandi Channel passage plan at Appendix A, issue 2 and current, names the "Primary means of position fixing: chart plotter" and the "Secondary means: visual transit of the No. 4 beacon and the town wharf leading lights". It lists the No. 4 beacon at "range 6 M". It closes: "this plan is written for the daylight transit. **Night transits are not scheduled on this route.**"

That note is the plan's only reference to darkness. There is no night passage plan for the Merrandi Channel, no guidance on what to do if the plotter is lost, and no abort or hold point anywhere in the plan. The only provision for slowing or stopping, at 3.6, applies "In reduced visibility", which this night was not.

Both masters interviewed said independently that the right response to a frozen plotter was to stop, and that nothing tells them to. Given the 114 second window before the vessel crossed the channel edge and the conditions described above, a named hold point would have applied throughout at no cost. Against this, a master who did not think of stopping unprompted might not have applied a written rule under the same pressure. The investigation accepted that possibility but found that it did not outweigh the length of the window available.

It is virtually certain that there was no night passage plan and no hold point for the loss of the primary means of position fixing, and very likely that this contributed to the occurrence by leaving the master with no prompt to stop when the plotter froze.

### The defect procedure and cleared faults

OPS-9 governs defect reporting:

- "9.1 A defect which renders the vessel unseaworthy or which affects a statutory item of equipment shall be reported to the Operations Manager immediately and entered in the vessel defect book." [Operations Manual OPS-9]

- "9.2 All other defects shall be entered in the vessel defect book at the end of the shift in which they are found, and will be reviewed by the Operations Manager weekly."  
[Operations Manual OPS-9]
- "9.3 At a change of master the outgoing master will brief the relieving master on the state of the vessel, any defect, and anything else affecting the conduct of the service."  
[Operations Manual OPS-9]

The chart plotter is not a statutory item on a sheltered-waters passenger vessel on this route, although it is the primary means of position fixing in the passage plan. An intermittent fault therefore falls under 9.2. For a fault that has cleared by the end of the shift, the provision gives no trigger, and the manual nowhere says whether such a fault counts as a defect. Section 9.3 does not require the brief to be written down, or the relieving master to confirm that a defect mentioned has been entered.

Both masters described the same reading independently: "The book is for things that are broken when you hand over. If it's broken you write it, if it came good you say it." The defect book bears this out. Its eleven entries in the 90 days before the occurrence all concern faults present when written.

The evidence that this procedure contributed is a history of freezes that never reached paper. On the outgoing master's account, the plotter had frozen three or four times in the preceding year. It had frozen on the relieving master a month earlier, and twice on the morning of 07/07/2026. None was written down, because each had cleared itself. Entered, they would have reached the weekly review and the maintenance system, and from there potentially the contractor, who identified cause code 0x51 against the 11/2024 firmware in a single bench test. In practice, the maintenance system held no work order, fault report or correspondence about the plotter at any time in 2026.

The evidence against is that the path from an entry to an applied fix runs through somebody ashore finding the manufacturer's bulletin, and the operator held no subscription to those bulletins. A recorded freeze might have been reviewed, noted and left. Nothing establishes how many recorded freezes it would have taken for the unit to be sent to the contractor, or how quickly.

The investigation weighed these points. A year of recorded freezes on the primary means of position fixing is unlikely to have gone unnoticed by everyone, and a single bench test was enough to identify the fix. How quickly the fix would have followed cannot be established.

It is virtually certain that the defect procedure did not require a fault that had cleared itself to be recorded, and likely that this contributed to the occurrence, because it kept the plotter's recurring freezes out of the maintenance system that could have led to the firmware fix.

## Organisation

### The move of the last service into darkness

The route risk assessment for the Merrandi Channel, RRA-07 issue 2 (03/2023), is the current issue. No issue 3 exists. It states:

- "Hazards assessed: Boolarong Bank to the west of the channel; channel width 40 m against a vessel beam of 8.2 m; single unlit reach between the No. 2 and No. 4 beacons; tidal range 1.4 m; recreational traffic at weekends." [RRA-07 issue 2]
- "Operating restriction: daylight transits only. A night transit of the Merrandi Channel requires the General Manager Operations' approval and a specific passage plan." [RRA-07 issue 2]

CHG-2026-014 moved TF18 from 16:15 to 17:35 for the winter timetable starting 01/06/2026. It recorded:

- "Assessment: timetable change only. Existing route, existing vessel, existing crew qualifications, existing berths. No new hazard identified." [CHG-2026-014]
- "Route risk assessment review required: No." [CHG-2026-014]
- "Approved: Operations Manager, 22/04/2026." [CHG-2026-014]

CHG-2026-014 does not refer to the daylight-only restriction. RRA-07 was not reopened and no night passage plan was written. The General Manager Operations' approval was neither sought nor given, and the change was approved one level below the person the restriction names. From 01/06/2026, TF18 crossed the channel after the end of civil twilight on every day it ran, and the occurrence transit was the thirty-first such crossing.

A review against a night operation would have reached the unlit reach between the No. 2 and No. 4 beacons, which the assessment already lists, and under the restriction could not have closed without a night passage plan. The change process instead asked whether the change introduced anything new, rather than whether the existing assessment still held.

Two points weigh against this factor. A reopened assessment might have produced a night plan no better than the daylight plan, with no hold point. And the Operations Manager gave his account of what he understood the restriction to require only verbally, not in a formal interview, so the investigation did not establish why the change was assessed as a timetable change only. The first point goes to how good a night plan would have been, not to whether one would have existed. A night plan written for this channel is the document most likely to have said what to do when the primary position fix was lost.

It is virtually certain that the last service was moved into darkness without reopening the daylight-only risk assessment, and likely that this contributed to the occurrence by leaving the night transit without the night passage plan that the assessment required.

### **The deferred clearing of the beacon's sightline**

Work order WO-118, raised in 09/2023, recorded mangrove regrowth on Merrandi Islet screening the No. 4 beacon from the northern approach, between about half a mile and a quarter mile. It recommended trimming to 2 m and a three-yearly cycle thereafter, at an estimated \$18,500.

The register shows the item deferred, unfunded and referred to the next round at each of the 2023, 2024 and 2025 works rounds. WO-176, raised in 03/2026, recorded the sightline as fully closed on the northern approach. The Authority's file note against WO-176 explains why the work kept losing out. The islet is 240 m from the beacon, outside the vegetation scope of the services agreement:

- "S2.4 Vegetation management within 50 m of an aid is a works item and is funded from the general waterways maintenance vote. Vegetation beyond 50 m is not within the scope of this agreement." [Services agreement, schedule 2]

The clearing therefore competed with boat ramps and pontoons in the general waterways vote rather than with other aids to navigation work.

The Authority's monthly check did not detect the problem. The agreement provides:

- "S2.1 The Authority will inspect each lit aid by daylight once per quarter and will confirm each lit aid's operation by observation from the town wharf once per month." [Services agreement, schedule 2]

The town wharf is 2.1 nm south of the beacon, and Merrandi Islet is not in that line of sight. The beacon was logged as operating at every monthly check from 01/2024 to 07/2026, the last on 02/07/2026.

The link between the deferral and the screening rests on the Authority's own figures. The 2019 aerial survey gives the canopy height as 2.1 m, and it measured 5.8 m in 07/2026, about half a

metre a year. A trim to 2 m funded at the 2023 round would have left the canopy near 3.5 m on the night, about 0.7 m below the 4.2 m lantern rather than 1.6 m above it, and the beacon would have been open over the stretch where it was screened. That calculation assumes even growth over the seven years, from two measurements and the work order's description.

It is virtually certain that the clearing of the beacon's sightline was deferred unfunded for three works rounds, and very likely that this contributed to the occurrence by allowing the canopy to grow above the lantern.

### The solar conversion of the No. 4 beacon

In 11/2025 the No. 4 beacon was converted from a mains supply to a solar lantern under the Council's waterways energy programme. The replacement is rated at a nominal 3 nautical miles, against 6 for the lantern it replaced. The light's character did not change.

A notification was lodged with the hydrographic service on 04/02/2026, and the chart correction was still pending on 07/07/2026. No local notice to mariners was issued and no operator was told. The investigation measured a luminous range of 2.6 nm on 18/07/2026. The operator's passage plan still lists the beacon at 6 M.

The investigation considered whether the reduced output affected the search for the light.

- **For.** The master called the beacon "a nothing" and said "It used to be a proper light. It was a mains light." For at least the last 37 seconds of his search the light was clear of the islet and not yet seen, and a weaker light against a bright background is harder to pick up.
- **Against.** Over the half mile in which the vessel left the channel, the canopy screened the beacon and a brighter light would not have shown through it. The master already knew the light was weaker, so a notice would have told him nothing new. Range was not limiting, since the search ran from 0.47 nm to 0.25 nm, well inside the measured 2.6 nm. Nothing suggests the plan's 6 M figure was used on the night.

Whether the light was conspicuous enough against the floodlights after it cleared the islet belongs with the floodlights, and remains unresolved there.

The change is still significant. A light on a 96-passenger route had its nominal range halved, the chart correction was pending eight months later, and no channel user was told. On a night when the light is not screened, or with another aid changed in the same way, a master working to the plan would be looking for a light twice as strong as the one there.

It is virtually certain that the beacon's nominal range was halved by the solar conversion without notice to channel users. It is unlikely that this contributed to the occurrence, and likely that it increased risk. This is a safety issue.

### The aids to navigation code

The Waterways Aids to Navigation Code WAN-2 sets the duties on the authority responsible for an aid:

- "4.1 The authority responsible for an aid to navigation must maintain it so that it performs the function for which it was established." [WAN-2 section 4]
- "4.2 Where an aid to navigation is established on a declared shipping channel, the responsible authority must record its availability, must achieve an availability of not less than 99.0 per cent measured annually, and must notify a change to the aid's character, position or range by notice to mariners before the change takes effect." [WAN-2 section 4]

- "4.3 Where an aid to navigation is established on a waterway that is not a declared shipping channel, section 4.2 does not apply. The responsible authority must be able to demonstrate that the aid is inspected at intervals it has determined and recorded."  
[WAN-2 section 4]

The Merrandi Channel is not a declared shipping channel, so section 4.2 does not apply to any of its six beacons. The services agreement adds no notification duty:

- "S2.7 ... There is no requirement in this agreement to notify waterway users of a defect, a deferral or a change to an aid." [Services agreement, schedule 2]

Section 4.1 applies to every aid. The Authority reads "performs the function for which it was established" as meaning lit, on station and showing its charted character, and on that reading considers itself compliant. The code does not define the function, or say whether a beacon that cannot be seen from part of the channel it marks is performing it.

Had section 4.2 applied, the range change would have gone out as a notice to mariners before it took effect, and the beacon's availability would have been recorded. Neither mechanism reaches this occurrence. A notice would have concerned a light screened where the vessel left the channel, and an availability figure measures whether a light is lit, which this one was throughout. Nor was the screening left in place because of the code, since section 4.1 applied regardless and the deferral of WO-118 arose from the 50 m vegetation scope.

The code still leaves a gap that matters. On a channel carrying a 96-passenger ferry in darkness, it gives the operator no right to be told when a beacon's range, character or position changes, and no figure for how often the aids work. That gap is why the solar conversion reached nobody for eight months, and the next change, to a light that is not screened, would reach nobody either. The Authority can close this gap through its own practice, whatever the code requires.

It is virtually certain that the code set no notice or availability duty for beacons on channels that are not declared shipping channels. It is unlikely that this contributed to the occurrence, and likely that it increased risk. This is a safety issue.

### **The winter roster**

In the winter timetable, TF18 closes the afternoon half of a split duty of 05:40 to 08:40 and 13:00 to 19:20. The master on that line works the same split on every day of a 12 days on, 2 days off tour. The duty totals 9 hours 20 minutes across a spread of 13 hours 40 minutes, against the operator's limits of 10 hours of duty and 14 hours of spread.

The darkness transit therefore falls in the last hour and a quarter of a spread that ends 20 minutes inside the limit, on days nine to twelve of the tour as on every other day. No relief master is rostered. A master who reports unfit is covered by extending another master's tour. On 07/07/2026 the master was on day nine, and had made 26 of the 30 darkness transits before the occurrence.

A roster can reach this occurrence only through the master's fatigue, which remains unresolved. His recorded performance shows no effect of fatigue on the night, and his failure to stop is matched by both masters' statements that no document asks that of them. The outgoing master, who had worked a morning shift, gave the same account of the handover and of the beacon.

The roster is nonetheless a standing exposure. The operator asks masters to keep a fatigue diary but does not collect it, and there is no relief master to break a tour.

It is virtually certain that the winter roster placed the darkness transit at the close of a split duty within 20 minutes of the spread limit. It is unlikely that this contributed to the occurrence, and likely that it increased risk. This is a safety issue.

## Hypotheses tested and excluded

Two further hypotheses were tested and ruled out, both by measurement.

**Tide set-down.** The investigation tested whether the tide had fallen below prediction over Boolarong Bank. It is very unlikely that it had. The Port Authority's gauge at Kanooka Heads read 0.91 m at 18:12, against a predicted 0.94 m, and its largest departure from prediction in the twelve hours to 20:00 was 0.07 m. With the bed at the contact point at 0.50 m below chart datum, there was 1.41 m of water over it. The vessel's effective draught was 1.53 m, made up of a static draught of 1.35 m and a calculated squat of 0.18 m. At the predicted height there would have been 1.44 m of water, so the vessel would have touched at the predicted tide as well. It grounded because it was 54 m outside a channel dredged to 3.2 m.

**Steering or propulsion failure.** The investigation tested whether a steering or propulsion failure occurred before the vessel left the centreline. It is very unlikely that one did, on two independent grounds:

- The engine recorder shows both engines at 1,780 and 1,784 rpm from 18:03 to 18:11:52, with no alarm, no excursion and no throttle movement. Steering hydraulic pressure held at 96 to 99 bar, within its normal band, and the rudder followed helm demand throughout. The rate of turn rose to a peak and returned to zero, the signature of a heading applied and then held rather than a sheer.
- The whole steering system was removed and function tested on 12/07/2026. Every component was serviceable, with no sign of a pre-existing defect. The damage to the starboard shaft, propeller and rudder dates from the contact.

## Findings summary

A finding stated without a likelihood term was established as fact.

### Contributing factors

- It was virtually certain that the last service, TF18, was moved into darkness without the daylight-only route risk assessment RRA-07 being reopened, and likely that this contributed to the grounding. (Safety Issue)
- It was virtually certain that there was no night passage plan for the Merrandi Channel and no hold point for the loss of the primary means of position fixing, and very likely that this absence contributed. (Safety Issue)
- The defect procedure in OPS-9 never required a fault that had cleared itself to be recorded. This was virtually certain to have existed and likely to have contributed. (Safety Issue)
- With virtual certainty, the chart plotter display froze during the channel transit at 18:07:06 and did not recover, and it was very likely that the freeze contributed.
- After the plotter froze, the master continued the transit at about eight knots until the shallow alarm sounded. This was virtually certain to have occurred and very likely to have contributed.
- The Port Authority deferred the clearing of the No. 4 beacon's sightline as unfunded at three successive works rounds, which was virtually certain to have occurred and very likely to have contributed. (Safety Issue)
- Mangroves on Merrandi Islet virtually certainly screened the No. 4 beacon from the channel between 0.84 nm and 0.34 nm north of it, and the screening likely contributed.
- The master eased the vessel's heading five degrees to starboard without a decision to turn, which was virtually certain to have occurred and virtually certain to have contributed.
- On the bank side, the vessel left the dredged channel, which was virtually certain to have occurred and virtually certain to have contributed.
- The echo sounder's shallow alarm, still at its 2019 commissioning setting of 0.9 m under the keel, gave 16 seconds' warning before contact. It was virtually certain that this occurred and very likely that it contributed.

### Other factors that increased risk

- The Waterways Aids to Navigation Code WAN-2 set no notice or availability duty for beacons off declared shipping channels. This was virtually certain to have existed and likely to have increased risk. (Safety Issue)
- In 11/2025 the No. 4 beacon's nominal range was halved by its conversion to a solar lantern, and no notice was given to users of the channel. This was virtually certain to have existed and likely to have increased risk. (Safety Issue)
- The winter roster placed the darkness transit at the close of a split duty whose spread ended 20 minutes inside the operator's limit, which was virtually certain to have existed and likely to have increased risk. (Safety Issue)
- A counter fitted in a 2019 refit without a re-survey of the safety equipment arrangement blocked the forward saloon lifejacket locker. It was virtually certain that this existed and very likely that it increased risk. (Safety Issue)

- At the change of master, the plotter fault was passed on verbally and was not entered in the defect book. This was virtually certain to have occurred and likely to have increased risk.

## Other findings

- A set-down of the tide below prediction over Boolarong Bank was examined and ruled out.
- A steering or propulsion failure before the vessel left the centreline was examined and ruled out.
- The engines were stopped and no attempt was made to refloat the vessel on a falling tide, which was examined and found to have reduced risk.

## Appendices

### Appendix A: Factor test summary

Each factor examined in the analysis was tested in three stages. The first stage asked whether the factor existed, and the second asked whether it influenced the occurrence. Where influence was not shown, a third stage asked whether the factor was important enough to address. The likelihood terms below are those recorded against each test. A factor whose existence or influence could not be settled is listed as unresolved, and it is not a finding.

| Factor                                                                               | Lane                      | Existence               | Influence               | Importance | Category                           |
|--------------------------------------------------------------------------------------|---------------------------|-------------------------|-------------------------|------------|------------------------------------|
| Last service moved into darkness without reopening the daylight-only risk assessment | Organisational Influences | Pass, virtually certain | Pass, likely            | Not tested | Contributing factor (Safety Issue) |
| Clearing of the beacon's sightline deferred unfunded for three works rounds          | Organisational Influences | Pass, virtually certain | Pass, very likely       | Not tested | Contributing factor (Safety Issue) |
| No night passage plan or hold point for loss of the primary fix                      | Risk Controls             | Pass, virtually certain | Pass, very likely       | Not tested | Contributing factor (Safety Issue) |
| Defect procedure never requires a fault that cleared itself to be recorded           | Risk Controls             | Pass, virtually certain | Pass, likely            | Not tested | Contributing factor (Safety Issue) |
| No. 4 beacon screened by mangroves from the channel, 0.84 to 0.34 nm                 | Local Conditions          | Pass, virtually certain | Pass, likely            | Not tested | Contributing factor                |
| Chart plotter display froze during the channel transit and did not recover           | Local Conditions          | Pass, virtually certain | Pass, extremely likely  | Not tested | Contributing factor                |
| Heading eased five degrees to starboard without a decision to turn                   | Individual Actions        | Pass, virtually certain | Pass, virtually certain | Not tested | Contributing factor                |
| Transit continued at eight knots from the plotter freeze until the shallow alarm     | Individual Actions        | Pass, virtually certain | Pass, extremely likely  | Not tested | Contributing factor                |
| Vessel left the dredged channel on the bank side                                     | Occurrence Events         | Pass, virtually certain | Pass, virtually certain | Not tested | Contributing factor                |
| Shallow alarm at its 2019 setting of 0.9 m gave sixteen seconds' warning             | Occurrence Events         | Pass, virtually certain | Pass, extremely likely  | Not tested | Contributing factor                |

|                                                                                             |                           |                                      |                                      |                   |                                                 |
|---------------------------------------------------------------------------------------------|---------------------------|--------------------------------------|--------------------------------------|-------------------|-------------------------------------------------|
| Aids code set no notice or availability duty for beacons off declared channels              | External Influences       | Pass, virtually certain              | Fail, unlikely                       | Pass, likely      | Other factor that increased risk (Safety Issue) |
| Beacon's range halved by solar conversion with no notice to channel users                   | Organisational Influences | Pass, virtually certain              | Fail, unlikely                       | Pass, likely      | Other factor that increased risk (Safety Issue) |
| Darkness transit rostered to close a split duty within 20 minutes of the spread limit       | Organisational Influences | Pass, virtually certain              | Fail, unlikely                       | Pass, likely      | Other factor that increased risk (Safety Issue) |
| Forward saloon lifejacket locker blocked by a 2019 counter fitted without re-survey         | Risk Controls             | Pass, virtually certain              | Fail, unlikely                       | Pass, very likely | Other factor that increased risk (Safety Issue) |
| Plotter fault passed on verbally at the change of master and not entered in the defect book | Individual Actions        | Pass, virtually certain              | Fail, unlikely                       | Pass, likely      | Other factor that increased risk                |
| Engines stopped and no attempt made to refloat on a falling tide                            | Individual Actions        | Pass, virtually certain              | Pass, very likely (decrease in risk) | Not tested        | Other finding (positive event)                  |
| Unshielded boat ramp floodlights aligned behind the No. 4 beacon from the north             | Local Conditions          | Pass, virtually certain              | Inconclusive, about as likely as not | Not tested        | Unresolved                                      |
| Master fatigued during the channel transit                                                  | Local Conditions          | Inconclusive, about as likely as not | Not tested                           | Not tested        | Unresolved                                      |
| Tide set down below prediction over Boolarong Bank                                          | Local Conditions          | Fail, very unlikely                  | Not tested                           | Not tested        | Excluded hypothesis                             |
| Steering or propulsion failure before the vessel left the centreline                        | Occurrence Events         | Fail, very unlikely                  | Not tested                           | Not tested        | Excluded hypothesis                             |

## Appendix B: Excluded hypotheses

### Tide set down below prediction over Boolarong Bank

The vessel grounded three hours into a falling tide, after an offshore westerly had blown for two days, so this hypothesis was tested. A set-down of a few tenths of a metre would have mattered. The existence of a set-down was assessed as very unlikely, however, and the hypothesis was excluded on measurement.

At 18:12 the Port Authority's gauge at Kanooka Heads read 0.91 m. The predicted height for that minute was 0.94 m, which put the tide 0.03 m below the tide table. That difference is the ordinary

scatter between a gauge and a table rather than a set-down. In the twelve hours to 20:00, the gauge's largest departure from the table was 0.07 m.

The bed at the contact point was surveyed at 0.50 m below chart datum, leaving 1.41 m of water over it. Against this, the vessel's static draught was 1.35 m and its calculated squat at 6.8 kt was 0.18 m, giving an effective draught of 1.53 m. At the predicted height there would have been 1.44 m of water, so the vessel would have touched at the predicted tide as well. It grounded because it was 54 m outside a channel dredged to 3.2 m. The water was not lower than the chart and the tide table said it would be.

### **Steering or propulsion failure before the vessel left the centreline**

Seen from outside the vessel, a five degree heading change followed by a steady set out of the channel is what a jammed rudder or a lost engine looks like. This hypothesis was therefore tested. Its existence was assessed as very unlikely, and it was excluded on two independent grounds.

The first ground was the recorded data. From 18:03 to 18:11:52 both engines ran at 1,780 and 1,784 rpm, within 12 rpm of each other, with no alarm, no excursion and no throttle movement. Steering gear hydraulic pressure held continuously at 96 to 99 bar, against a normal band of 90 to 105 bar. Throughout, rudder angle followed helm demand within 1.5 degrees. The rate of turn peaked at 6 degrees per minute and then returned to zero. That pattern is a heading applied and then held, not a sheer.

The second ground was physical examination. On 12/07/2026 the whole steering system was removed and function tested. Every component was serviceable, with no internal leakage and no sign of a pre-existing defect. The damage to the starboard shaft, propeller and rudder all dated from the contact, not from before it. In addition, the bridge audio contained no remark about steering or the engines at any point.

## **Appendix C: Sources of evidence**

### **Interviews and accounts**

- The master was interviewed on 14/07/2026, and the interview was recorded. He also submitted safety report SR-2026-0288, timed 18:52 on the day of the occurrence.
- The deckhand was interviewed on 14/07/2026, and at her request the interview was not recorded. She also submitted safety report SR-2026-0291.
- The outgoing master was interviewed by telephone on 23/07/2026. The interview was not recorded, but the notes were read back to him and agreed.
- The Operations Manager gave a verbal account of how he understood the daylight restriction in RRA-07. It was not taken as a formal interview.
- The master's fatigue diary gave his sleep on the night before the occurrence.

### **Captured material**

- Port Control AIS receiver log and vessel recorder readout AIS-2026-0093-R1, issued 15/07/2026. This covered the AIS track, the engine monitoring recorder and the steering data for service TF18.
- AUD-2026-0093, the VHF channels 12 and 16 and bridge camera audio from 17:32 to 19:26 local. It was transcribed on 16/07/2026 and verified by the Port Authority's traffic officer.
- The chart plotter's internal event log, recovered 09/07/2026, and the contractor's bench test report of 21/07/2026.

- The echo sounder setup and alarm log, recovered 09/07/2026.
- MET-TIDE-0093-070726, retrieved 09/07/2026. This held the tide predictions, the gauge readings, the weather observations, and the light and moon data.
- The grounding position survey and bed levels, surveyed on 08/07/2026.
- The survey and slipping report for MV Tarrakin Star, 09/07/2026 to 14/07/2026, including the steering gear examination.
- The stowage survey of the MV Tarrakin Star forward saloon, 16/07/2026.
- The range and visibility check of the No. 4 beacon, made from the port workboat on the inbound track on 18/07/2026.
- The Kanooka Heads foreshore lighting survey, made from the port workboat on 18/07/2026.
- The Kanooka Heads Port Authority aids to navigation register and register extract for the No. 4 beacon, copied 19/07/2026, including work orders WO-118 and WO-176.
- The Kanooka Shire Council and Kanooka Heads Port Authority aids to navigation services agreement, schedule 2.
- The Waterways Aids to Navigation Code WAN-2, section 4.
- Tarrakin Ferries Operations Manual OPS-3 revision 6, 02/2023, and appendix A issue 2, the Merrandi Channel passage plan.
- Tarrakin Ferries Operations Manual OPS-9 revision 4, 08/2024, covering defect reporting and rectification.
- Route risk assessment RRA-07 issue 2 (03/2023) and change record CHG-2026-014 (approved 22/04/2026).
- Tarrakin Ferries crew records, certificates and the roster for 07/07/2026, copied 13/07/2026.
- The wheelhouse defect book and maintenance system extract for MV Tarrakin Star, copied 10/07/2026.

## Appendix D: Investigation activity

- 07/07/2026: The investigation was notified by the master's own safety report SR-2026-0288, which arrived before the operator's notification was raised. The occurrence was assessed as a serious marine incident, and an investigation was opened.
- 08/07/2026: The vessel had refloated overnight and was secured at the town wharf. The plotter and echo sounder heads were removed and sealed. The Port Control AIS log and the VHF recordings were requested, and the bridge camera was downloaded before slipping. The master and the deckhand were stood down and tested, with negative results. The General Manager Operations suspended the evening Muntari services, and the grounding position survey was carried out the same day.
- 10/07/2026: The grounding position survey and bed levels were received, and the steering gear was removed for function testing. The deckhand's safety report SR-2026-0291 was also received. On the strength of it, the forward saloon lifejacket stowage was added to the scope.
- 14/07/2026: The master and the deckhand were interviewed, and both raised the No. 4 beacon. A night range and visibility check of the beacon from the port workboat, on the inbound track, was then requested.
- 18/07/2026 to 19/07/2026: The range and visibility check found that the beacon is not visible from the channel between 0.84 nm and 0.34 nm north of it. It also established that

the solar conversion of 11/2025 had halved the beacon's nominal range without notice to any operator. The foreshore lighting was surveyed the same night. The aids to navigation register was also obtained, and it showed WO-118 deferred at three works rounds.

- 23/07/2026: The outgoing master was interviewed by telephone. Both masters had raised the beacon unprompted, yet neither had ever reported it. The company's 'navigation hazard' reporting category was found to be unused on this route and unknown to all of the crew interviewed. These matters were raised with the Port Authority and the operator.

## Distribution

| Title                                             | Name         |
|---------------------------------------------------|--------------|
| General Manager Operations                        | H. Vardon    |
| Safety Manager                                    | P. Nandasiri |
| Operations Manager                                | C. Bickerton |
| Harbour Master, Kanooka Heads                     | L. Truscott  |
| Aids to Navigation Officer, Kanooka Shire Council | E. Maguire   |